CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (CBIT)

Department Of Computer Science And Engineering

FUNDAMENTALS OF DATA SCIENCE (22CSC59)



CASE STUDY REPORT

Title: Wine Quality Assessment

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1. Problem statement and objective

The company currently relies on sensory quality assessments. This case study investigates whether laboratory measurements can help explain differences in quality scores and support a data-driven screening process. The tasks are descriptive statistics, exploratory visualizations, correlation analysis, multiple linear regression, and business.

Dataset and audit

Item Finding

Red wine records 1,599

White wine records 4,898

Combined records 6,497

Predictor features 11 physicochemical measurements

Missing cells 0

Repeated rows ignoring wine type 1,179

Target Sensory quality score

The red and white files are combined for descriptive analysis. The regression uses exactly the 11 physicochemical

predictors specified in the case brief; wine_type is retained only to audit the source composition and is not a model

predictor. Repeated records are retained rather than arbitrarily deleted. The pooled dataset contains many more white

wine records than red wine records, which affects the overall summaries.

Methodology

- Calculated mean, median, mode, standard deviation, skewness, and coefficient of variation (CV = SD / mean) for alcohol, volatile acidity, and residual sugar.
- Examined Pearson correlation, quality-score distribution, alcohol box plots grouped by quality, and a correlation heatmap.
- Used an 80/20 train-test split with random_state = 42. StandardScaler was fitted on training features only; ordinary multiple linear regression was evaluated on the held-out test set.
- Reported MAE, MSE, RMSE, R-squared, standardized predictor coefficients, pairwise correlations, and variance inflation factors (VIF).

2. **Task 1 - Statistical findings**

Descriptive statistics for the three required attributes

Attribute Mean Median Mode Std. dev. Skewness CV

alcohol 10.4918 10.3000 9.5 1.1927 0.566 11.4%

volatile acidity 0.3397 0.2900 0.28 0.1646 1.495 48.5%

residual sugar 5.4432 3.0000 2.0 4.7578 1.435 87.4%

Relative variability. Residual sugar has the highest coefficient of variation (87.4%), followed by volatile acidity

(48.5%) and alcohol (11.4%). Residual sugar therefore varies the most relative to its mean among these three

variables. Volatile acidity and residual sugar have positive skewness, meaning a longer right tail; their means exceed

their medians.

Alcohol and quality. Pearson r = 0.4443, indicating a moderate positive linear association in the pooled red + white

sample. Higher alcohol content tends to coincide with higher quality scores, but the association is far from deterministic

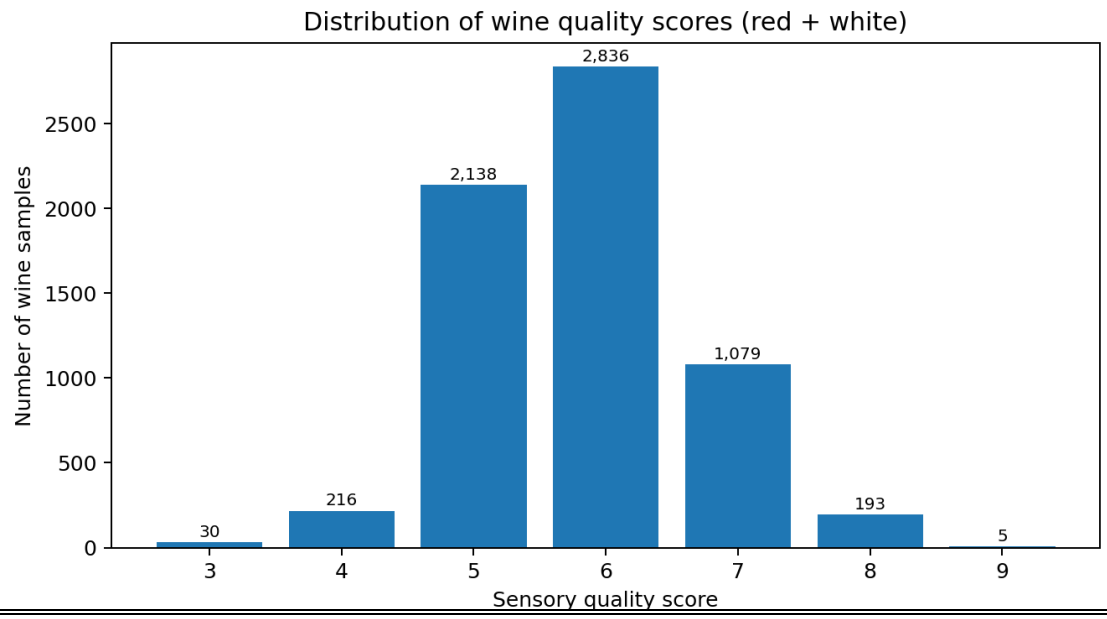
and does not establish causality.

3. Task 2 - Exploratory data analysis

Figure 1. Quality scores are concentrated at 5 and 6: 2,138 samples score 5 and 2,836 score 6. Scores 3, 8 and 9 are rare, and no

score outside 3-9 appears in the supplied combined data. The imbalanced distribution makes extreme quality scores harder to

model.



4. Exploratory patterns and correlation

Figure 2. The median alcohol content generally rises across higher quality scores, consistent with the positive Pearson correlation.

The boxes overlap, however, so alcohol alone cannot reliably distinguish the score of an individual wine. Points beyond the whiskers

indicate potential outlying measurements within a quality group; they are not automatically data errors.

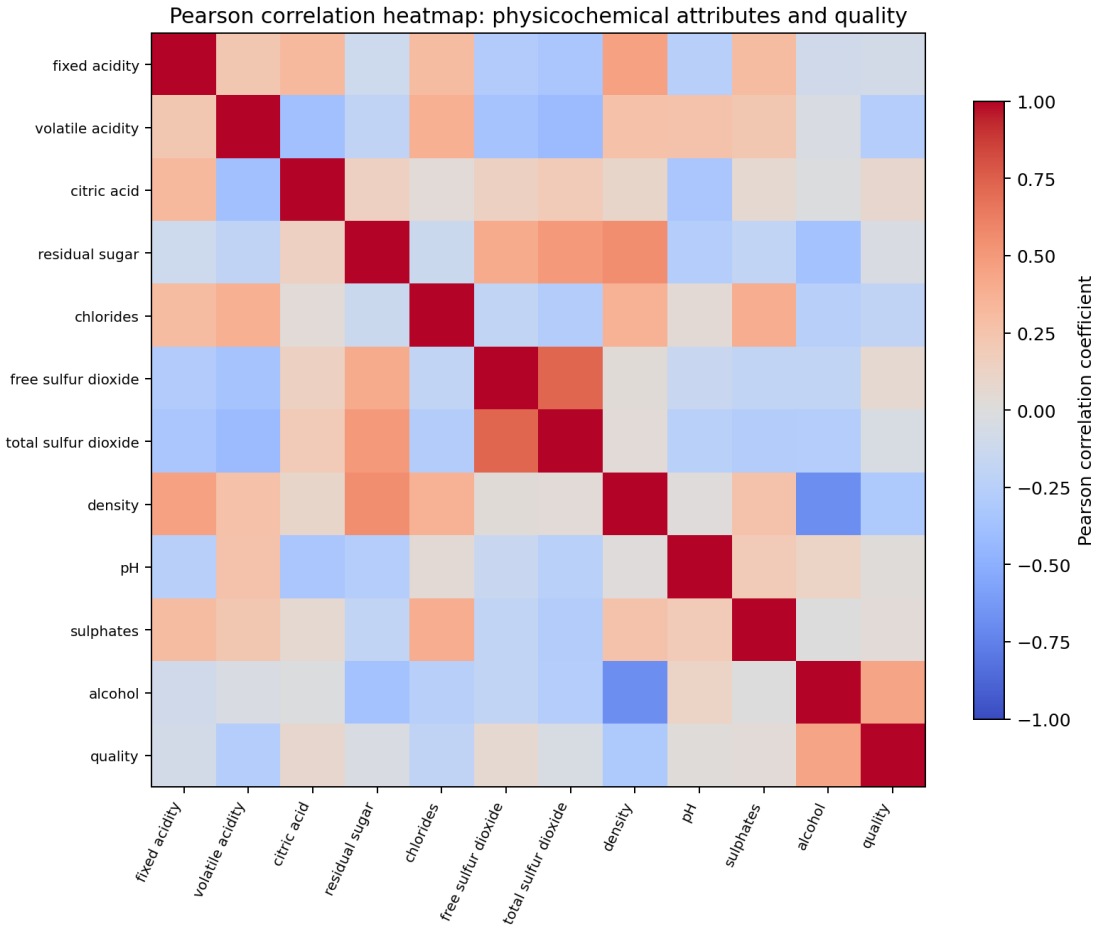
Correlation heatmap and two strongest associations

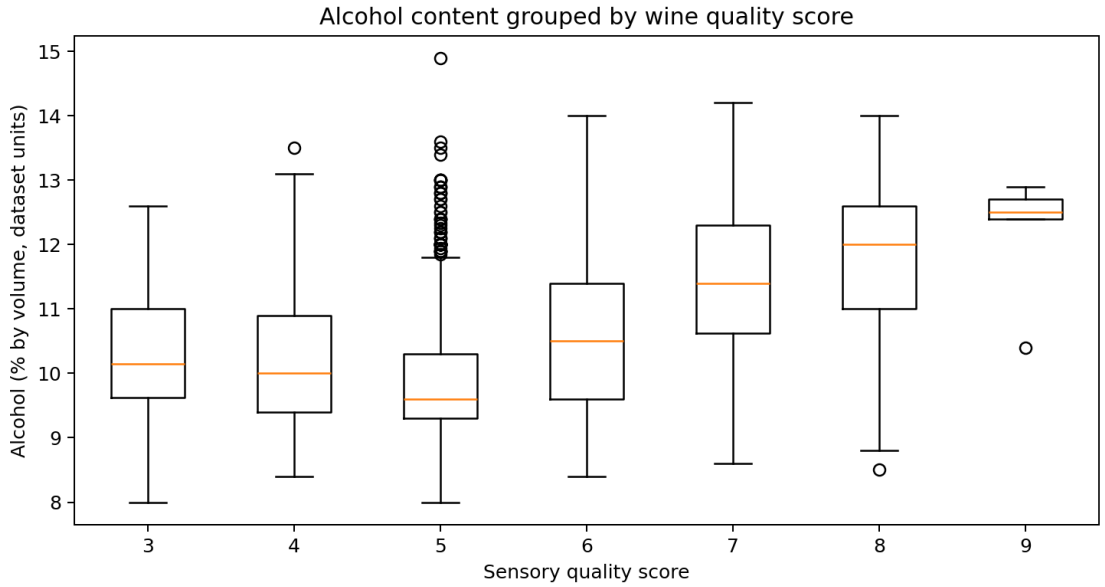
Figure 3. Among the 11 physicochemical predictors, alcohol has the strongest absolute Pearson correlation with quality (r = +0.444).

Density is second (r = -0.306). These are moderate-to-weak associations rather than near-perfect relationships. Density is also

correlated with other predictors, including alcohol (r = -0.684 in the training data) and residual sugar (r = +0.563), so its marginal

correlation should not be interpreted in isolation.





5. Task 3 - Regression model evaluation

A multiple linear regression model was fit with all 11 physicochemical variables. The split produced 5,197 training

records and 1,300 test records; `random_state = 42` was fixed for reproducibility. Predictors were standardized using

the training set before model fitting.

Held-out test metrics

Metric Test value Interpretation

MAE 0.5659 Average absolute prediction error is about 0.57 quality points.

MSE 0.5467 Mean squared error; larger errors receive greater penalty.

RMSE 0.7394 Typical error scale is about 0.74 quality points.

R-squared 0.2598 The model explains about 26.0% of test-set quality variance.

The test R-squared of 0.260 indicates limited predictive power. An MAE of about 0.566 is below one full quality point

on average, but the score range is discrete and high/low grades are rare; average error does not guarantee reliable

decisions on individual samples.

Standardized regression coefficients

Rank Attribute Standardized coefficient

1 alcohol +0.3224

2 residual sugar +0.2204

3 volatile acidity -0.2190

4 density -0.1773

5 total sulfur dioxide -0.1495

6 free sulfur dioxide +0.1225

7 sulphates +0.1202

Alcohol is the largest positive standardized coefficient (+0.322). Residual sugar (+0.220) and volatile acidity (-0.219) have similar

absolute coefficient sizes but opposite signs after adjustment for the other model inputs. These coefficients describe conditional

linear associations, not causal effects.

Multicollinearity diagnostic

Predictor VIF Pair of predictors Pearson r

density 15.51 free sulfur dioxide / total sulfur dioxide +0.724

residual sugar 7.20 density / alcohol -0.684

fixed acidity 4.74 residual sugar / density +0.563

alcohol 4.62 residual sugar / total sulfur dioxide +0.492

total sulfur dioxide 2.96

Density has a VIF of about 15.51, and residual sugar has a VIF of about 7.20, indicating substantial overlap with other

predictors. The largest pairwise correlation shown is between free and total sulfur dioxide (r = +0.724); density and

alcohol are also correlated (r = -0.684). Multicollinearity can make coefficient magnitudes and signs less stable even

when overall predictions remain usable.

6. Prediction quality and business decision-making

Figure 4. Test-set actual versus predicted scores. Actual test scores range from 3 to 9; predictions range from 4.05 to 7.72.

Predictions cluster around the middle of the range and do not reach the rare observed extremes, consistent with the moderate

MAE/RMSE and low R-squared. A linear model tends to regress uncommon scores toward the center.

Business recommendation

Alcohol and density deserve attention during screening because they have the two strongest absolute Pearson

correlations with quality in the pooled data. Density should be interpreted alongside other measurements due to its

multicollinearity with alcohol and residual sugar. Screening thresholds should be validated against sensory scores

before the company uses them for quality decisions.

Can the model replace sensory evaluation?

No. The test-set R-squared of 0.260 shows that most variation in sensory quality remains unexplained, and the

prediction scatter shows limited reproduction of rare high and low scores. The model can prioritize samples for review

or serve as an assistive signal, but replacing trained sensory evaluation is not supported by these results.

Limitations

- The red and white wine datasets are pooled, but wine type is not included as a predictor as requested; the larger

white-wine group dominates pooled statistics.

- Quality scores are imbalanced; only 5 samples have score 9, and extreme scores are poorly represented.
- Linear regression cannot capture nonlinear relationships or interactions unless explicitly engineered.
- Density and other predictors are correlated, which limits isolated interpretation of coefficients.

Conclusion

Alcohol is moderately positively associated with quality, density is moderately negatively associated, and residual

sugar has the greatest relative variability among the three required attributes. The linear regression has moderate

average error but explains only about one quarter of test-set variance. Laboratory measures can support triage and

monitoring; sensory evaluation remains necessary for final quality decisions.

Dataset citation supplied in winequality.names: Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. (2009). Modeling wine

preferences by data mining from physicochemical properties. Decision Support Systems, 47(4), 547-553. DOI: 10.1016/j.dss.2009.05.016.

Execution url: https://colab.research.google.com/drive/1KeZ5ztC0eKBQCzJ6iWKDfdcCg1LNATSH?usp=sharing